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Financial Statement Analysis

Financial statement analysis involves the use of simple mathematical techniques, an understanding of accounting and an appreciation of business' strategy to gain insights into the reporting corporation's history, current position and future prospects through an examination of the corporation's financial statement. The results of these analyses can play an important role in credit decisions, valuing securities, analyzing competitors, and appraising managerial performance. This note introduces some of the basic tools of financial statement analysis. A knowledge of these techniques is essential to appreciate fully the communication aspect of financial reports. The valuation and rating of debt securities and its related financial analysis techniques are discussed in a later note. In addition, many of the ratios discussed in this note are also covered in greater detail in other notes dealing with accounting issues and transactions that are the specific focus of a ratio.

A Business Analysis

Competent financial statement analysts approach their task as if it was a business analysis. Accordingly, before beginning a financial statement analysis they first identify the reporting company's strategy, predict the possible consequences of this strategy, and assess whether it is appropriate for this company.

This initial step is useful for several reasons. First, it helps the analyst avoid the problem of "not seeing the forest for the trees." That is, it reduces the probability that the analyst will become focused to such an extent on the details of the analysis that a larger and more critical view of the company is missed. Second, it gives the analysis a focus that facilitates the forming of relevant interpretations of the data and analytical results. Third, it provides a framework for presenting the results of the financial statement analysis in a meaningful and coherent communication.

Strategy identification and analysis is a complex task that is beyond the scope or focus of this note. Readers should look elsewhere for an understanding of this topic, but look they should because an understanding of alternative strategies models and the optimum strategies under a given set of conditions should help analysts to appreciate better the significance of the historical, comparative and projected financial statement statistics developed by their analysis. For example, in some instances the financial analysis may indicate that the company has adopted the optimum strategy for its circumstances. This revelation may explain why the trends and ratios are what they are. It may provide also some clues as to what the future strategy of the company might be, which would, in turn, imply certain trends and ratios. Alternatively, the analysis might indicate that the optimum

Professor David Hawkins prepared this note as the basis for class discussion.

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strategy is not being followed. This could raise questions as to why the optimum strategy is not being followed, which may result in the identification of some unique characteristics of the company that suggest the theoretical best strategy is not appropriate. Or, if the company indeed is following a suboptimizing strategy, a knowledge of the theoretical consequences of this action may improve the analyst's ability to forecast the consequences of the actual strategy.

Financial Reporting Bias

A company's financial reporting decisions influence its financial statement data. Accordingly, before subjecting financial statement data to any ratio or other type of detailed analysis, experienced analysts assess where and how management's financial reporting decisions may have biased the financial statement data. In particular, the analyst examines the financial statements and accompanying notes looking for business transactions and management accounting principle elections, account caption decisions, format choices and measurement judgements that might bias the data or the analyst's perception of the data. If the analyst concludes the financial statement data has been materially biased by management, the analyst may attempt to eliminate the bias by adjusting the raw data.

Financial Ratio Analysis

Financial ratios are the fundamental analytical tools for interpreting financial statements. Financial ratio analysis mathematically relates items in the financial statements in a meaningful manner. The analyst evaluates these results against the particular characteristics of the company and its industry. The astute analyst seldom expects answers from this process. Rather, the analyst hopes it will provide clues as to where to focus subsequent analysis that may involve such investigation techniques as company visits, supplier and customer interviews, library searches, and credit bureau report reviews.

Financial ratios fall into four classes: ratios appraising *liquidity*, ratios measuring *solvency*, ratios evaluating *funds management*, and ratios measuring *profitability*. The categories indicate that different ratios may be more helpful than others for particular purposes. Therefore, rather than calculating ratios indiscriminately, the experienced analyst begins by considering the kinds of insights that will be helpful in understanding the problem faced. The analyst then calculates those ratios that best serve his or her purpose. To get the most meaningful results, the analyst compares these ratios over a period of several years against some standard; examines in depth major variations from this standard; and cross checks the various ratios against each other.

The balance sheet and income statement of The Ampex Corporation, a retailing business, will be used to illustrate some of the more common ratios (see **Illustrations A and B**).

Illustration A Ampex Corporation, Comparative Balance Sheets,
December 31, 2011, and 2012 (in thousands of dollars)

Assets	2012	2011
Current assets:		
Cash	\$ 20	\$ 30
Accounts receivable (net)	95	95
Inventory	<u>130</u>	<u>110</u>
Total current assets	<u>\$ 245</u>	<u>\$ 235</u>
Fixed assets:		
Land	\$ 10	\$ 10
Building and equipment (net)	<u>120</u>	<u>100</u>
Total fixed assets (net)	<u>\$130</u>	<u>\$110</u>
Other assets:		
Goodwill and organization costs	<u>\$ 10</u>	<u>\$ 10</u>
Total assets	<u>\$385</u>	<u>\$355</u>
Liabilities and Stockholder's Equity		
Current liabilities:		
Accounts payable	\$ 50	\$ 40
Estimated income taxes payable	<u>10</u>	<u>10</u>
Total current liabilities	\$ 60	\$ 50
Fixed liabilities:		
Mortgage bonds, 10 percent	<u>\$ 50</u>	<u>\$ 50</u>
Total liabilities	<u>\$110</u>	<u>\$100</u>
Stockholders' equity:		
Convertible preferred stock, 5 percent	\$ 20	\$ 20
Common stock (10,000 shares outstanding)	50	50
Retained earnings	<u>205</u>	<u>185</u>
Total stockholders' equity	<u>\$275</u>	<u>\$255</u>
Total liabilities and stockholders' equity	<u>\$385</u>	<u>\$355</u>

Illustration B Ampex Corporation Condensed Income Statement, 2012
(in thousands of dollars)

Gross sales	\$11,516	100.66%
Less: Returns and allowances	<u>10</u>	<u>0.66</u>
Net sales	\$ 1,506	100.00%
Less: Cost of goods sold	<u>1,004</u>	<u>66.67</u>
Gross profit	\$ 502	33.33%
Operating expenses ^a	<u>400</u>	<u>26.56</u>
Operating profit	\$ 102	6.77%
Interest	<u>5</u>	<u>0.33</u>
Profit before taxes	97	6.44%
Income tax expense	<u>47</u>	<u>3.12</u>
Net income	\$ 50	3.32%
Less: Preferred dividends	1	0.07
Common dividends	<u>29</u>	<u>1.93</u>
Change in retained earnings	<u>\$ 20</u>	<u>1.32%</u>

^aIncludes lease rental costs of \$30,000 and depreciation of \$10,000.

Liquidity Ratios

A corporation's liquidity is measured by its ability to raise cash from all sources, such as bank credit, sale of redundant assets, and operations. Liquidity ratios have a narrower focus. They help statement users appraise a company's ability to meet its current obligations using its existing cash and current assets. These ratios compare current liabilities, which are the obligations falling due in the next 12 months, and current assets, which typically provide the funds to extinguish these obligations. The difference between current assets and current liabilities is called "working capital." (Ampex's 2012 year-end working capital is \$185,000 (\$245,000 – \$60,000).)

Current Ratio

$$\frac{\text{Current assets}}{\text{Current liabilities}} = \frac{\$245,000}{\$60,000} = 4.1$$

The meaningfulness of the current ratio as a measure of liquidity varies from company to company. Typically, it is assumed that the higher the ratio, the more protection the company has against liquidity problems. However, the ratio may be distorted by seasonal influences, slow-moving inventories built up out of proportion to market opportunities, or abnormal payment of accounts payable just prior to the balance sheet date. Also, the nature of some businesses is such that they have a steady, predictable cash inflow and outflow, and a low current ratio is appropriate for such a business.

Acid-Test or Quick Ratio

$$\frac{\text{Quick assets}}{\text{Current liabilities}} = \frac{\$115,000}{\$60,000} = 1.9$$

The acid-test or quick ratio measures the ability of a company to use its "near-cash" or quick assets to immediately extinguish its current liabilities. Quick assets include those current assets that presumably can be quickly converted to cash at close to their book value. Such items are cash, stock investments, and accounts receivable. Like the working capital ratio, this ratio implies a liquidation approach and does not recognize the revolving nature of current assets and liabilities.

Solvency Ratios

Solvency ratios generate insight into a company's ability to meet long-term debt payment schedules. There are a number of ratios that compare stockholders' equity or operating profits to the amount and cost of funds provided by creditors. All of these ratios are designed to give some measure of the extent to which operating cash flows and asset values provide protection to creditors should a company incur losses.

Times Interest Earned

$$\frac{\text{Operating profit (before interest expense)}}{\text{Long-term debt interest}} = \frac{\$102,000}{\$5,000} = 20.4 \text{ times}$$

This coverage ratio is calculated on a pretax basis, since bond interest is a tax deductible expense. The ratio in the example implies that operating profits cover interest payments 20 times. This

indicates the extent to which operating profits can decline without impairing the company's ability to pay the interest on its long-term debt.

Some analysts prefer to use operating profit plus noncash charges as the numerator of this ratio. This modification indicates the ability of the company to cover its cash outflow for interest from its funds from operations. For example, the only so-called noncash charge in the Ampex income statement is depreciation; that is, no cash outflow results from incurring this expense. Adding the company's \$10,000 depreciation expense to operating profit changes the numerator to \$112,000 and increases the coverage to 22 times.

Coverage ratios can be computed for preferred stock dividends and other fixed charges, such as lease rentals. The *preferred-stock-dividend-coverage* ratio is calculated on an after-tax basis, since preferred stock dividends are not a tax deductible expense. For example, the Ampex preferred-stock-dividend-coverage ratio is:

$$\frac{\text{Net income}}{\text{Preferred stock dividends}} = \frac{\$50,000}{\$1,000} = 50 \text{ times}$$

A coverage ratio for all of a company's fixed charges is called the *times-fixed-charges-earned* ratio. The denominator of this ratio includes such items as lease rentals, interest, and preferred dividends converted to a pretax basis. The numerator is operating profit before these charges. The times-fixed-charges-earned ratio for Ampex is:

$$\frac{\text{Operating profit before fixed charges}}{\text{Lease rentals, interest, preferred dividends}} = \frac{\$132,000}{\$34,000} = 3.9 \text{ times}$$

Sometimes depreciation is added to the numerator.

Debt-to-equity ratios The relationship of borrowed funds to ownership funds is an important solvency ratio. Capital from debt and other creditor sources is more risky for a company than equity capital. Debt capital requires fixed interest payments on specific dates and eventual repayment. If payments to a company's creditors become overdue, the creditors can take legal action which may lead to the company being declared bankrupt. Ownership capital is less risky. Dividends are paid at the discretion of the directors, and there is no provision for repayment of capital to stockholders. It is generally assumed that the more ownership capital relative to a debt a company has in its capital structure, the more likely it is that the company will be able to survive a downturn in business that may force other more financially-leveraged companies into bankruptcy. An excessive amount of ownership capital relative to debt capital may not necessarily indicate sound management practices, however. Equity capital is typically more costly than debt capital. Also, the company may be foregoing opportunities "to trade on its equity," that is, borrow at a relatively low interest rate and earn a greater rate of return on these funds. The difference between these two rates can increase earnings per share without having to increase the number of common shares outstanding.

There are a number of debt-to-equity ratios. Four of the most common are:

$$\frac{\text{Total liabilities}}{\text{Total assets}} = \frac{\$110,000}{\$385,000} = 28.6\% \text{ or } 0.286 \text{ to } 1$$

This ratio indicates the proportion of a company's total assets financed by short- and long-term credit sources.

$$\frac{\text{Long-term debt}}{\text{Capitalization}} = \frac{\$50,000}{\$325,000} = 15.4\% \text{ or } 0.154 \text{ to } 1$$

This measure, which excludes current liabilities, reflects management's policy on the mix of long-term funds obtained from ownership and nonownership sources. The term "capitalization" includes all of a company's long-term debt and equity capital.

$$\frac{\text{Total liabilities}}{\text{Stockholders' equity}} = \frac{\$110,000}{\$275,000} = 40\% \text{ or } 0.4 \text{ to } 1$$

This ratio is another way of measuring the relative mix of funds provided by owners and creditors.

$$\frac{\text{Total liabilities}}{\text{Stockholders' equity}} = \frac{\$385,000}{\$275,000} = 1.4$$

The *total leverage* ratio relates total assets to stockholders' equity. This ratio is an indication of the degree to which management has financed the company's asset investments with nonownership capital. A 1.4 ratio means that for every dollar of owners' equity, \$0.40 of nonownership funds has been used to fund the company's assets.

Ampex appears to have an adequate cushion of ownership funds against losses from operations, decreases in the book value of assets, and downturns in future cash flows.

Funds Management Ratios

The financial situation of a company turns in large measure on how its investments in accounts receivable, inventories, and fixed assets is managed. As a business expands its sales, it is not uncommon to find that the associated expansion of these three items is so great that despite profitable operations the company is short of cash.

Receivables to Sales

$$\frac{\text{Accounts receivable (net)}}{\text{Net sales}} = \frac{\$95,000}{\$1,506,000} = 6.3\%$$

In the absence of an aging of accounts receivable (classification of outstanding receivables by days since billing) or other detailed credit information, the receivables-to-sales ratio, computed over a number of years, can give a crude indication of the trend in a company's credit policy. In those cases where a company sells for cash and credit, only credit sales should be used in the denominator. Receivables include accounts receivable, trade receivables, and trade notes receivable. The rather low receivables-to-sales percentage for Ampex is indicative that this retailer's sales most probably include a high proportion of cash sales.

Average Collection Period

$$\frac{\text{Accounts receivable}}{\text{Net sales}} \times \text{Days in the annual period} = \text{Collection period}$$

$$6.3\% \times 365 \text{ days} = 23 \text{ days}$$

A two-step method to get the same result is:

1. Calculate the average daily sales:

$$\frac{\text{Net sales}}{\text{Days}} = \frac{\$1,506,000}{365 \text{ days}} = \$4,126 \text{ per day}$$

2. Calculate the day's sales represented by receivables.

$$\frac{\text{Accounts receivable}}{\text{Net sales per day}} = \frac{\$95,000}{\$4,126} = 23 \text{ days}$$

To appraise the quality of accounts receivable, the average collection period can be related to the typical credit terms of the company and its industry. A collection period substantially longer than either of these standards might indicate credit management problems, resulting in an increasing amount of funds being tied up in this asset. On the other hand, a significantly shorter collection period than is typical in the industry might mean that profitable sales to slower paying customers are being missed.

Average Accounts Payable Period

$$\frac{\text{Accounts payable}}{\text{Purchases}} = \frac{\$50,000}{\$1,024,000} = 4.9\%$$

Similar tests can be made of accounts payable to see how well they are managed. In this case, the accounts payable are compared to the purchases for the period (cost of goods sold plus inventory changes). The two-step calculation of the average day's payables is made as follows:

1. Calculate the average daily purchases:

$$\frac{\text{Purchases}}{\text{Days}} = \frac{\$1,004,000 + \$20,000}{365 \text{ days}} = \$2,805 \text{ per day}$$

2. Calculate the day's purchases represented by payables:

$$\frac{\text{Accounts payable}}{\text{Purchases per day}} = \frac{\$50,000}{\$2,805} = 18 \text{ days}$$

The day's-payables ratio becomes meaningful when compared to the credit terms given by suppliers to the object company's industry. If a company's average day's payables is increasing, it may mean trade credit is being used increasingly as a source of funds. If the payables period is less than the average for the industry, it may indicate that management is not using this source of funds as much as possible. If it is longer, it may mean the company is overdue on its payables and is using this source of funds beyond the normal trade limits.

Rarely is the purchases figure available to people outside of the company. Consequently, the analyst has to approximate this amount. In merchandising situations, like the Ampex illustration, estimating purchases is fairly straightforward since a retailer's inventory value and cost of goods sold is the price paid to suppliers for the goods. Therefore, a retailer's outside purchases is equal to its cost of goods sold figure, adjusted for inventory changes.

In manufacturing situations, estimating outside purchases is not such an easy task. The cost of goods sold expense includes direct labor, raw materials, and some manufacturing overheads. If the

raw materials portion of the cost of goods sold and inventories figures are available, they can be used to calculate an approximation of the raw materials purchases, which in most cases represents the minimum level of purchases.

Another difficulty is that the accounts payable figure may include payables incurred for other than items included in cost of goods sold. As a result of this problem and the other measurement problems, this ratio is usually not regarded as a particularly reliable indicator of the use of trade credit.

Inventory Turnover

$$\frac{\text{Cost of sales}}{\text{Average inventory}} = \frac{\$1,004,000}{\$120,000} = 8.4 \text{ times}$$

The inventory-turnover ratio indicates how fast inventory items move through a business. It is an indication of how well the funds invested in inventory are being managed. The analyst is interested in two items: the absolute size of the inventory in relation to the other funds needs of the company and the relationship of the inventory to the sales volume it supports. A decrease in the turnover rate indicates that the absolute size of the inventory relative to sales is increasing. This can be a warning signal, since funds may be tied up in this inventory beyond the level required by the sales volume, which may be rising or falling.

Average inventory is used in the denominator because the sales volume is generated over a 12-month period. The average inventory is obtained by adding the opening inventory and closing inventory balances and dividing the sum by two.

If the cost of goods sold figure is not available, an approximation of the inventory-turnover rate can be obtained by using the sales figure in the numerator. If profit margins have remained fairly steady, then this sales-to-average-inventory ratio can provide, over a period of years, an indication of inventory management trends.

By dividing the turnover rate into 365 days, the analyst can estimate the average length of time items spent in inventory:

$$\frac{365 \text{ days}}{\text{Inventory turnover}} = \frac{365}{8.4} = 43 \text{ days}$$

If the company uses the so-called last-in, first-out (LIFO) inventory valuation method, the inventory balance must be converted to its equivalent value based on either the first-in, first-out (FIFO) or average cost inventory valuation method. This is accomplished by adding the LIFO reserve to the FIFO inventory balance reported on the balance sheet. The LIFO reserve figure can be found in the inventory note accompanying the financial statements.

Asset Turnover

$$\frac{\text{Net sales}}{\text{Average assets}} = \frac{\$1,506,000}{\$370,000} = 4.07 \text{ times}$$

The asset turnover ratio is an indicator of how efficiently management is using its investment in total assets to generate sales. High turnover rates suggest efficient asset management.

Fixed Asset Turnover

$$\frac{\text{Net sales}}{\text{Average net fixed assets}} = \frac{\$1,506,000}{\$120,000} = 12.6 \text{ times}$$

A similar turnover ratio can also be calculated for fixed assets. It provides a crude measure of how well the investment in plant and equipment is being managed relative to the sales volume it supports. The usefulness of this measure is reduced considerably because book values seldom approximate market values or are comparable from company to company due to different depreciation policies.

Capital or Investment Turnover

$$\frac{\text{Net sales}}{\text{Average total capital}} = \frac{\$1,506,000}{\$315,000} = 47.8 \text{ times}$$

A ratio similar to the inventory and asset turnover ratios can also be calculated for a firm's total capital investment.

Profitability Ratios

Analysts look at profits in two ways: first, as a percentage of net sales; second, as a return on the funds invested in the business.

Profit margin Profit margins relative to net sales can be evaluated in a number of different ways.

Net Margin

$$\frac{\text{Net income}}{\text{Net sales}} = \frac{\$50,000}{\$1,506,000} = 3.32\%$$

Net income as a percentage of net sales measures the total operating and financial ability of management, since net profit after taxes includes all of the operating and financial costs of doing business.

Pretax Margin

$$\frac{\text{Pretax margin}}{\text{Net sales}} = \frac{\$97,000}{\$1,506,000} = 6.44\%$$

Pretax income as a percentage of net sales measures management's ability to generate profits before recognizing taxes related to those profits. This figure is of interest because changes in the tax code or actions by management that impact the company's tax rate can result in more or less of the pretax profits flowing down to net income.

EBIT

$$\frac{\text{Earnings before interest and taxes}}{\text{Net sales}} = \frac{\$102,000}{\$1,506,000} = 6.77\%$$

The ratio of net profit before taxes and interest to net sales is indicative of management's operating ability. Interest is excluded because it relates to financing policy rather than operation efficiency.

Gross Margin

$$\frac{\text{Gross profit}}{\text{Net sales}} = \frac{\$502,000}{\$1,506,000} = 33.3\%$$

Gross profit (sales minus cost of sales) as a percentage of sales is an indication of the management's ability to mark up its products over their cost.

In addition to these ratios, it is often informative to express all of these expense items as a percentage of net sales (see *Illustration B*). This is called a common-size statement. Common-size statements can also be prepared for balance sheets using total assets as the base.

Return on investment The relationship between profitability and investment is considered the key ratio by many analysts. It provides a broad measure of management's operating and financial success. Several different return-on-investment ratios are commonly used.

Pretax Operating Return on Total Assets

$$\frac{\text{Profit before taxes and interest}}{\text{Average total assets}} = \frac{\$102,000}{\$370,000} = 27.6\%$$

This ratio gauges how well management has managed the total resources at its command, before consideration of taxes and credit costs. It focuses on the earning power of the assets and is not influenced by how they are financed. Average total assets is used as the denominator since profit is earned over a 12-month period.

Return on Total Assets

$$\frac{\text{Net income}}{\text{Average total assets}} = \frac{\$50,000}{\$370,000} = 13.5\%$$

This variation measures the return on total assets after recognition of taxes and financing costs.

Return on Total Capital

$$\frac{\text{Net income}}{\text{Average total capital}} = \frac{\$50,000}{\$315,000} = 15.9\%$$

Another ratio measuring return on investment equates investment with total long-term capital (equity capital plus long-term liabilities). This ratio indicates how well management has managed the permanent funds at its disposal. The ratio can be computed on a before- or after-tax basis. If interest is paid on long-term liabilities, then this amount is sometimes added to the net income figures, since it relates to the financial management of those items.

Return on Stockholders' Equity

$$\frac{\text{Net income}}{\text{Average stockholders' equity}} = \frac{\$50,000}{\$265,000} = 18.9\%$$

The return on net worth percentage measures the return on ownership capital after all taxes and interest payments. It is perhaps the most common return-on-investment figure published by financial services.

Return on Tangible Net Worth

$$\frac{\text{Net income}}{(\text{Average net worth} - \text{Average intangible assets})} = \frac{\$50,000}{(\$265,000 - \$10,000)} = 19.6\%$$

This modification measures the return on net worth less the intangible assets, such as goodwill and capitalized organization costs. The principal use of this ratio is to present a more conservative measure of the investment base. The \$10,000 deduction from net worth in the above equation is Ampex's investment in goodwill.

Linking Ratios

Statement users can often gain greater insight into a company's returns on capital, assets, and net worth by linking together selected financial ratios.

For example, how a company achieves its return on total capital can be explained in terms of its capital turnover and net margin by using the following equation:

$$\text{Capital turnover} \times \text{After-tax profit margin} = \text{Return on capital}$$

$$\frac{\text{Net sales}}{\text{Average total capital}} \times \frac{\text{Net income}}{\text{Net sales}} = \frac{\text{Net income}}{\text{Average total capital}}$$

$$4.78 \text{ times} \quad \times \quad 3.32\% \quad = \quad 15.95\%$$

This formula indicates that a business return on investment can be improved by increasing the sales volume per dollar of capital, by generating more net income per dollar of sales, or some mix of these two factors. Thus, the performance of a firm earning 2% on sales with a capital turnover of 10 times can be equivalent to that of another company with a profit margin of 10% and an investment turnover of 2 times. Both have the same return on investment, 20%.

A number of other ratios are closely related. For example, a greater appreciation of the relationship between profitability, asset turnover, and return on total assets can be obtained as follows:

After-tax profit margin $\times$ Asset turnover = Return on total assets

$$\frac{\text{Net income}}{\text{Net sales}} \times \frac{\text{Net sales}}{\text{Average total assets}} = \frac{\text{Net income}}{\text{Average total assets}}$$

$$3.32\% \quad \times \quad 4.07 \text{ times} \quad = \quad 13.5\%$$

An analysis of the causes of changes in the level and quality of the company's return on stockholders' equity can be facilitated by the use of the equation presented in *Illustration C*, which incorporates the Ampex Corporation data and ratios. The average, rather than ending, stockholders' equity and total asset values are used to compute the total leverage ratio because the equation is being used to explain the variables contributing to the company's return on average stockholders' equity.

The only new ratio introduced in *Illustration C* is the so-called *tax retention* ratio, which measures the percentage of pretax profits retained by the company after payment of income taxes. This ratio is expressed as 1 minus the book tax rate, which is the percentage of the book tax expense to the pretax profits. The equation demonstrated in *Illustration C* is often referred to as the Du Pont equation after the name of the company that first used it extensively to analyze the performance of its divisions, subsidiaries and investee corporations.

A similar set of ratios which have more meaning when examined together are the components of the current ratio. For example, an examination of the relationship between the inventory turnover period and the receivables and payables periods demonstrates how changes in these working capital items influence a company's working capital financing requirements.

Illustration C Analysis of Ampex Corporation's Return on Stockholders' Equity

Pretax profit margin	$\times$	Asset turnover ratio	$\times$	Total leverage ratio	$\times$	Tax retention rate	=	Return on stockholders' equity
$\frac{\text{Pretax profits}}{\text{Net sales}}$	$\times$	$\frac{\text{Net sales}}{\text{Average assets}}$	$\times$	$\frac{\text{Average assets}}{\text{Average stockholders' equity}}$	$\times$	(1-Tax rate)	=	$\frac{\text{Net Income}}{\text{Average stockholders' equity}}$
6.44%	$\times$	4.07	$\times$	1.40	$\times$	(1 - .48)	=	19.1%
Pretax return on assets (6.44% $\times$ 4.07 = 26.2%) Pretax return on stockholders' equity (6.44% $\times$ 4.07 $\times$ 1.40 = 36.7%) Return on stockholders' equity (6.44% $\times$ 4.07 $\times$ 1.40 $\times$.52 = 19.1%)								

Some refinements In practice, there are a number of minor refinements to the basic Du Pont equation.

Some users of the expanded equation prefer to use long-term capital, rather than total liabilities, in the turnover and leverage ratios on the grounds that they are interested in how a company uses its “permanent” capital. If capital is defined as long-term liabilities plus owners’ equity, the ratios used in the Du Pont equation are:

$$\text{Turnover} = \frac{\text{Sales}}{\text{Capital}}$$

$$\text{Leverage} = \frac{\text{Capital}}{\text{Owners' equity}}$$

A few analysts use the pretax income plus interest charges, rather than income before income taxes, as the numerator in the margin ratio. When a company’s interest charges are large relative to pretax income, these analysts are interested in what is described as “pure” operating margin figure and “pure” return on assets ratio. The ratios to adjust the Du Pont equation are:

$$\text{Pretax margin} = \frac{(\text{Pretax income} + \text{interest})}{\text{Sales}}$$

$$\text{Return on assets} = \frac{(\text{Pretax income} + \text{interest})}{\text{Assets}}$$

To convert the pretax margin before interest deduction to the margin based on the reported pretax income figure, the following ratio has to be inserted into the expanded equation after the pretax before interest margin:

$$\text{Ratio to adjust pretax before interest margin to pretax margin} = \frac{\text{Pretax income}}{(\text{Pretax income} + \text{interest})}$$

A Warning

Financial ratio analysis has many limitations. It can easily mislead the unsophisticated analyst. To be a useful analytical tool it must be used wisely.

First, ratio analysis deals only with quantitative data. It does not look at qualitative factors such as management’s ethical values, the quality of the management, or the workers’ morale. These are important considerations which should be taken into account when evaluating a company.

Second, management can take certain short-run actions prior to the statement dates to influence the ratios. For example, a company with a better than 1:1 current ratio can improve this ratio by paying off current liabilities just prior to the balance sheet date.

Third, comparison of ratios between companies can be misleading because of differences in accounting practices in such areas as depreciation, income recognition, and intangible assets. For this reason, analysts often put companies on a comparable accounting basis before making ratio comparisons.

Fourth, different definitions of common ratios are used by different analysts. This can lead to misleading comparisons and interpretations.

Fifth, because accounting records are maintained in historical dollars, a change in the purchasing power of the dollar due to inflation can distort the comparability of the ratios computed for different time periods. For instance, in periods of inflation the ratios comparing sales and net income to assets and equity may be biased upwards.

Sixth, a ratio standing alone has no significance. What constitutes an appropriate ratio for a company is determined by its industry, management strategy, and the state of the general economy. For example, meat packers have high inventory-turnover ratios and jewelry stores have low inventory-turnover ratios. To conclude one ratio is good and the other bad is a mistake. Ratios results must be evaluated in their business context.

Finally, ratios based on published financial statements show relationship as they existed in the past. The analyst interested in the future should not be misled into believing that the past data necessarily reflects the current situation or future expectations.

Common Stock Ratios

Financial statement analysis and equity valuation models incorporate a number of ratios relating the market value of common stocks to financial statement related items, such as earnings, net book value and dividends.

Earnings Per Share

The most straightforward computation of earnings per share is for companies with fairly simple capital structures, like Ampex. In these situations, the calculation is:

$$\frac{(\text{Net income} - \text{Preferred stock dividends})}{\text{Average number of common shares outstanding}} = \frac{\$50,000 - \$1,000}{10,000} = \$4.90 \text{ per share}$$

This earnings per share figure is known as “basic earnings per share.”

Preferred stock dividends, if any, are deducted from net income before calculating basic earnings per share because this portion of the net income is not available to common stockholders. The denominator is the weighted average number of shares determined by the portion of time within a reporting period common shares have been outstanding to the total time in that period.

The number of earnings-per-share figures a company must report will vary with the complexity of its capital structure. The accounting rules for computing earnings per share require that companies with complex capital structures present with equal prominence two types of earnings-per-share amounts on the face of the income statement: one, basic earnings per share; the other, diluted earnings per share. Basic earnings per share is the amount of earnings attributable to each share of common stock outstanding. Diluted earnings per share is the amount of current earnings per share reflecting the maximum dilution that would result from conversions and exercises, of convertible debt, warrants, options and other issues with contingent rights to common stock that individually, in the future, may decrease earnings per share and in the aggregate might have a dilutive effect. It is

always dangerous to use earnings-per-share figures without knowing which definition is being used.¹

Typically, financial analysts eliminate nonrecurrent items from a single-year analysis of companies. In single-year analysis, financial analysts tend to want to know whether or not the current earnings are in line with the “normal” or continuing earnings of the company. Consequently, extraordinary items, discontinued operations, adjustments for new accounting principle adoptions, and unusual items such as material refunds of overpaid taxes are excluded from these analyses.

In long-run analyses of historical data, financial analysts tend to include in income every profit and loss item, unless it is quite unrelated to normal operations. This practice recognizes that many of these so-called unusual items are elements of profit and loss that would have been included in income if the accounting period had been longer than, say, one year. In this latter case, the tax refund excluded from the single-year analysis would be included. An example of items typically excluded from long-term analyses would be gains from pension plan terminations.

Price-Earnings Ratio

Assuming the price for the Ampex stock is \$40, the company’s price-earnings ratio is:

$$\frac{\text{Market price per share}}{\text{Earnings per share}} = \frac{\$40.00}{\$4.90} = 8.2\times$$

Since the stock market is responsive to anticipated earnings-per-share estimates, the price-earnings ratio is often quoted using the company’s projected next year’s earnings-per-share figure. Typically, the net income figure before extraordinary items is used by analysts to compute this ratio. Also, as noted above, the potential shares issuable to holders of potentially dilutive securities is included in the earnings-per-share calculation.

The reciprocal of the price-earnings ratio gives the so-called *capitalization rate* applied by investors to the company’s earnings per share.

$$\frac{\text{Earnings per share}}{\text{Market price per share}} = \frac{\$4.90}{\$40.00} = 12.2\%$$

Book Value per Share

Book value per share is the relationship between a company’s stock price and its net assets, expressed on a per common share basis. Assuming the Ampex’s stock price is \$40, the company’s price to book value ratio is:

$$\frac{\text{Market price per share}}{\text{Net assets per common share}} = \frac{\$40.00}{\$25.50} = 1.57\times$$

The book value per common share ratio excludes the preferred stock portion of stockholders’ equity.

¹The detailed rules for calculating earnings per share are covered in a later note.

Dividend Yield

Dividend yield is the relationship between a company's dividend and its stock price. In situations where interim cash dividends have been increased at the last payment date, the current dividend rate converted to a yearly basis is sometimes used for the numerator:

$$\frac{\text{Cash dividends per share}}{\text{Price per share}} = \frac{\$ 2.90}{\$40.00} = 7.3\%$$

Stock dividends are not included in this calculation. The *payout ratio* is the percentage of net income paid out in cash dividends:

$$\frac{\text{Cash dividends}}{\text{Net income}} = \frac{\$29.00}{\$50.00} = 58\%$$

Growth Analysis

Past and expected growth rates of a corporation's sales, profits, and dividends is a major focus of many financial statement analyses. Investors are interested because of the close relationship between equity stock values and the projected growth rate and the expected volatility of earnings and dividends. Creditors examine past growth records in order to predict the future level of funding required to finance changes in accounts receivables, inventories, and productive assets. Corporations scrutinize growth rates of themselves and their rivals in order to ascertain how well they are doing on a comparative basis, to detect potential weaknesses in their competitors, and to predict future competitive behavior.

Typically, the analysis of growth characteristics and growth rates involves three major steps: First, a quantitative measurement of the rate of growth of the variables being examined. This quantification may range from a simple index number series to a complex mathematical equation describing the trend pattern inherent in the data. Second, the identification of the various sources of the growth. This phase involves both quantitative and qualitative assessments of the sources of growth as well as an analysis of the relationship of these various sources to each other, the operating and financial characteristics of the firm, and the external environment during the period under review. Third, the use of the preceding analyses in combination with projections such as projected industry growth rates, management's announced capital expansion plans, other analysts' forecast of future levels of growth to develop forecast growth rates and to speculate on the possible financial and operating consequences of the projected growth rates.

Sustainable Growth Rate Equations

Historical and pro forma growth rate analyses use a convenient mathematical expression known as the sustainable growth rate equation. This equation demonstrates if a company does not issue new equity, its potential maximum earnings growth rate will be a function of its rate of return on equity and dividend payout policy.

For example, assume that a company had earnings of \$10 million during the year just ended; a net worth of \$100 million at the beginning of that year; and a permanent dividend payout policy of 50%. Thus, during the year just ended, the company earned 10% on its beginning net worth; retained \$5 million of earnings, and ended the year with an equity of \$105 million. If the past year's 10% return

on beginning equity is repeated during the next year, the company's earnings will grow to \$10.5 million, i.e., 10% of this year's beginning net worth of \$105 million. This 5% earnings growth rate will be repeated annually as long as the company continued to earn 10% on each year's beginning net worth and pays out 50% of its earnings in dividends.

The relationship between future earnings growth, return on beginning equity, and dividend payout policy described in this example is captured in more general terms in the sustainable growth rate equation

$$g = \text{ROE} \times \left(1 - \frac{D}{E}\right)$$

where:

g = Annual net income growth rate or sustainable growth rate.

ROE = Rate of return on beginning-of-year net worth, including preferred stock.

$\frac{D}{E}$ = Dividends payout ratio, i.e., annual common and preferred dividends (D) divided by annual earnings (E).

$\left(1 - \frac{D}{E}\right)$ = Earnings retention rate, i.e., 1-the dividend payout ratio.

By observation, it can be seen from the above equation that a future earnings growth rate cannot be greater than a company's return on beginning equity, i.e., $g = \text{ROE}$, when dividends are zero. In addition, should a company wish to maintain a given earnings growth rate, it can do so by a variety of combinations of payout ratio and return on equity levels. For example, should the company in the above example find that its projected return on beginning equity had slipped to, say 8%, it would have to adjust its payout ratio downward to 37.5% if it wished to maintain its 5% earnings growth rate in the future $[\text{.08} \times (1 - .375) = .05]$.

In practice, a company's actual earnings growth rate may differ for at least four reasons from the rate indicated by the sustainable growth rate calculation. The first two reasons are fairly obvious: the actual return on equity and the actual payout ratio may differ from the ones used in the equation. The other two reasons require some explanation.

The sustainable growth rate is a maximum potential rate of growth. If a company's industry is growing slower than the company's potential growth rate and the company cannot increase its market share, then the industry growth rate becomes the sustainable growth rate.

Also, as noted earlier, the sustainable growth rate computation assumes a company's only source of new equity is retained earnings. If new equity is sold, then a company's earnings in the year in which the equity is sold will be derived from both the equity retained in the prior year and the new capital. This can result in a one-year earnings growth that is greater than the original projected growth rate.

Analysis of Income Change

In contrast to the growth equation analyses, which focus on the operating and financial causes of the *rate of growth*, growth analysis also involves accounting for the sources of a company *dollar growth*

in earnings. This dollar oriented analysis develops answers to the question: “What accounted for the dollar change in earnings from one period to another?” The answer to this question is presented frequently in terms of the dollar contribution to the change in earnings of these six variables:

- Accounting.
- Environment.
- Infrequent and/or unusual transactions.
- Tax rate.
- Capital structure.
- Operations.

An accounting source of earnings change may arise from a change in an accounting principle estimate or the way in which a principle is implemented. For example, a decision to use longer depreciation lives in the current period could be a source of earnings growth over the prior period, since the prior period’s income will be based on the old depreciation schedule. The “Accounting Changes” notes to financial statements usually identifies and gives the dollar value of these sources of earnings change.

Environmental sources of earnings change include all of those domestic and international economic changes that influence earnings, other than interest rates. Typically these are: changes in the business cycle of the economy and/or the industries of the company, shifts in foreign exchange rates, and changes in the rate of inflation. It is difficult to relate net income changes to changes in the business cycle and/or the industries of a company. Often, the analyst has to be content to make such statements as: “Income improved due to an improvement in the economy.” However, no matter how crude the final figure related to this source of earnings change, the analyst should try to gauge the relative significance of this influence. The impact on earnings of shifts in foreign exchange rates are reported in the notes. This figure relates to the balance sheet related gain or loss. In addition, the impact of the exchange rate shift on the translated foreign currency profits of the foreign income sources must also be estimated. In the case of inventories of LIFO companies, the impact of inflation is captured in the size of the LIFO reserve adjustment to cost of goods sold. FIFO companies do not make such a debit or credit to income. In this case, this missing debit or credit is the inventory-inflation source of profit change. The extent of the impact on profit changes of using historical cost depreciation in periods of inflation may be crudely estimated by computing how much the historical depreciation of a company falls short of its replacement cost depreciation.

Infrequent and/or unusual transactions may occur in one period and not another and as such are a source of earnings change. Examples of such transactions are extraordinary items and infrequent or unusual items included in income before extraordinary items, such as write-down of inventory or the gain on the sale of equipment.

Tax rate sources of earnings change are usually reflected directly in a change in the tax rate between two periods. The current income before taxes times the prior period’s tax rate less the net income reported for the current period measures this influence. The tax expense note should provide clues as to what caused the company’s book tax rate to change.

Capital structure derived changes in income include such transactions as: the conversion of debt securities to common stock, which changes the interest charged to income, and the acquisition and extinguishment of a company’s own debt securities, which changes the interest expense and may generate a gain or loss on the transaction.

Operating sources of income changes are those changes that can be identified with the ongoing business of the company. These include such sources of income change as market share shifts, changes in the cost structure, different product mixes, and new product prices schedules.

Real Versus Nominal

Most financial trend data is expensed in nominal dollars. Nominal dollars reflect the purchasing power of the period. These data are not adjusted for changes in the purchasing power of the dollar during the period covered. It may be useful, particularly in high inflation rate periods or when data spanning more than five years is being examined, to adjust the nominal data for price level changes.²

Basis of Comparison

The results of financial statement analysis take on real meaning when compared to a standard appropriate to the company's stage of development, seasonal pattern, and industry and management plans. The selection of a relevant standard is always difficult.

The management of a company can use its budget as a basis of comparison. These are rarely available to the outside analyst. Therefore, the statement user must seek other sources. By comparing a company's current results as shown in its financial reports to similar data in past reports, the analyst can get some indication of how much "better" or "worse" things are compared to the past.

Important sources of average data for a particular industry are the investor reference services' publications in most countries. Another source of comparison bases are the publications of the various trade associations. Often, these publications report selected financial data for industries broken down by sales volume and asset categories. These data can be used to highlight variations from the typical company situation.

Another source of standards can be ratios computed from the data in the annual reports of companies in the same industry. This type of external comparison, when used with good judgment, can indicate the relative quality of the company's operating performance and financial management compared to its competitors.

Experienced financial analysts rarely rely on any one standard. They use several standards. They also look at a variety of related ratios and know from experience that they must have a good appreciation of the particular company's business before drawing conclusions based upon financial statement analysis.

Transnational Financial Statement Analysis

With the globalization of the world's economies, statement users are increasingly faced with the need to read, analyze and understand financial statements issued by companies located in many countries. This activity is known as transnational financial statement analysis. In this setting the task of the analyst is complicated because the analysts must deal with financial statements that are often based on unfamiliar accounting rules, expressed in a foreign language and currency, and present

²A later note discusses the various techniques for converting nominal dollar financial data to a real or price level adjusted basis.

financial data in an unfamiliar format. Underlying these differences may be cultural, tax and legal conditions peculiar to the reporting company's country that influence a company's attitudes toward financial reporting principle and disclosure decisions.

Formats

Statement formats can vary considerably from country to country. For example, some income statement formats show expenses by function (wages) while others classify expenses by type (production wages, selling wages, etc.). Some balance sheets show assets by increasing order of liquidity while others list assets in the reverse order. Classification of items within financial statements can also differ. Transnational financial analysts deal with the multi-format problem by either learning to work with different formats, recasting foreign financial statements to their domicile country's format or recasting all financial statements to a standard format developed by the analyst.

Language

Few statement users have the multi-language skills required to feel comfortable dealing with financial statements presented in languages other than their own. Some companies recognized this problem and publish their financial reports in several languages. In some cases statement users employ translators. Translations can be helpful, but often the translation is prepared by accountants who know accounting but are not good translators or by translators who are not accountants. In either case this can lead to confusing and misleading translations. Statement users must be alert to this problem. Other solutions include using accounting lexicons to translate foreign account terms or limiting the analysis to the financial statement data in standard investor reference books. These two solutions often result in the analyst using summary data without a full understanding of the company's true condition or the accounting policy decisions.

Different Terminology

Transnational financial analysts must become familiar with accounting terminology differences. Some of these differences are easy to deal with as one term can be directly substituted for the other. For example, turnover is the British equivalent for what is called sales in the United States of America. Other differences are more difficult to handle. For example, the word extraordinary in a Japanese financial report does mean the same thing as when it is used in a U.S. company's financial statements. In these cases, close reading of the financial statement is required to determine the exact meaning of the terms.

Currencies

Typically companies issue financial statements expressed in the currency of the parent company's country. Financial ratios are unaffected by the reporting currency, except in the case where a company for the convenience of foreign readers restates its reporting currency financial statements into other currencies. These so-called convenience statements are prepared by multiplying all items in the statements by the period end exchange rate. Analysts using these data must be careful to not be misled by distortions in period to period comparisons and trend analyses caused by changes in exchange rates. If the analysts wants to use convenience reports to analyze trend data, then the data must be scaled to eliminate the exchange rate effect. This can be done by using a consistent exchange rate over time. The best way to avoid being misled by convenience statements is to use the original

reports to calculate trend data. This approach has the added advantage of giving the analyst the same view of the reporting company as its management, who is most probably measuring itself and managing the company based on the original statements.

Accounting Principles

Despite the success of the movement to converge national accounting standards around the world, there is still considerable diversity of accounting standards around the world. For example, some countries, such as the Republic of Ireland, permit revaluation of property, others, such as the United States of America, insist on valuing it at its original purchase price, while others, such as Mexico, require during periods of high inflation that the cost of property be adjusted for inflation.

How these differences are handled in financial analyses depends on the intended use of the analysis and the analyst's comfort level in dealing with different accounting models and rules. If the analyst is analyzing different companies within a single country or a single company over time, in most cases a satisfactory financial ratio analysis can be prepared using the financial statements data as reported regardless of the accounting rules employed. Accounting diversity becomes a major problem in comparative analysis of financial statements from different countries with dissimilar accounting standards. In this situation the analyst usually recasts the financial statement data to make it comparable using a common accounting basis, such as the analyst's home country accounting standards or an accounting model of the analyst's own creation. Since restatement to a common basis is often difficult to accomplish and inevitably leads to at best rough approximations of comparable data, analysts tend to focus on reconciling only those accounting differences responsible for material differences in the original comparative data.

Disclosure

The extent of financial disclosure by corporations can vary considerably between countries and within some countries between listed and private companies. When confronted with the need to analyze a company with a low level of financial disclosure, such as a Swiss company, the analyst has little choice than to seek the missing information from noncompany sources, such as local investment bankers, competitors, retired employees and others familiar with the company. This will increase the financial analysis cost, but it may give the analyst a valuable advantage over less diligent analysts who are simply content to analyze the limited financial data provided by the company.

Tax Code

In some countries the tax code requires in a large number of instances identical accounting be used in financial statements if certain tax accounting rules are elected for tax reporting purposes. In other countries, companies are not bound except in a limited number of cases by such tax code requirements. Transnational analysts must be aware of situations where tax return accounting elections may have a significant distorting influence on financial statements. The financial statements may not reflect the economic realities of the reporting company, since the goal of tax management is usually to minimize or defer taxable income and tax accounting rules are adopted by governments to achieve revenue, social and economic objectives.

Legal

In many countries corporate reporting practices are governed by company legislation. Typically, this legislation provides some overarching requirements for financial reporting and, in some cases, may specify specific accounting practices.

Cultural

A country's accounting culture or traditions can influence the way in which local accounting standards are employed in practice. For example, German accounting has a long history of being credit oriented with the result that prudence is encouraged in financial reports. That is, assets tend to be understated, liabilities tend to be overstated and disclosure clearly limited. In contrast, United Kingdom accounting has for a long time been geared more to the information needs of public equity investors who want transparent financial reports that reflect the current economic realities of the reporting company. Given these different perspectives on accounting, it is very likely that a German company and a British company could apply the same accounting standard to an identical transaction and end up with very different accounting results. Transnational financial analysts need to be aware of these accounting cultural differences so that they do not fall into the trap of assuming the same accounting rule necessarily lead to comparable accounting data.

Prediction

Financial ratios, in particular combinations and weighting, are often used to estimate the probability of future financial outcomes, such as the probability that a particular loan applicant would default on a loan if granted. Typically, these multivariate prediction equations, as they are referred to, develop a score that prior research involving the testing of multiple combinations and weighting of a ratio set has indicated is best associated with particular outcomes.

The Altman Z-score is an example of the use of a multivariate prediction equation. It is a popular and reasonably reliable predictor of the probability of corporate bankruptcy.³ The Basic Altman Z-score formula is:

$$Z = 1.2T_1 + 1.4T_2 + 3.3T_3 + .6T_4 + .999T_5$$

where:

T_1 = Working Capital / Total Assets

T_2 = Retained Earnings / Total Assets

T_3 = Earnings Before Interest and Taxes / Total Assets

T_4 = Market Value of Equity / Book Value of Total Liabilities

T_5 = Sales / Total Assets

A Z-score greater than 2.99 indicates that a company is unlikely to go bankrupt; a score between 1.8 and 2.99 is a "grey" zone; and, a score of less than 1.80 indicates the probability of bankruptcy is very high within a 2-year period.

³ Edward I. Altman, "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy," *Journal of Finance*, September 1968.

The basic Altman Z-score formula was derived by comparing the financial ratio characteristics of a group of publicly-traded manufacturing companies that went bankrupt and a similar group of manufacturing companies that did not go bankrupt. The Altman Z-score proved to be a reasonably reliable discriminator between the two groups of companies.⁴

Efficient Markets

A branch of financial economics research known as efficient markets research has reached these following tentative conclusions that may be relevant to the field of financial analysis and equity valuation: The major equity markets are efficient in that security prices typically reflect all of the publicly available accounting and nonaccounting data. Inside information gives the holder of it an advantage over other investors until it is used. Financial statement data are used in investment decisions. The market is able to “look through” most accounting differences to distinguish real economic changes from the apparent changes reported by using different accounting alternatives. Finally, in the long run security markets reflect economic reality.

Efficient market researchers believe these preliminary findings may have a number of implications for the field of financial analysis. Since financial analysis techniques and its results are widely known, the value of using known analytical tools must already be impounded in security prices. Therefore, if financial analysis is to be beneficial, it must be helpful in predicting future results or have some novel quality. However, once these techniques and the data they produce become public, the advantages of the predictor and innovator will be quickly lost.

Another finding suggested by the research into the role of accounting data in equity markets is that companies using the more liberal accounting practices tend to have the higher market risk, as measured by the volatility of a stock’s price relative to some index of the volatility of the market as a whole. Since companies that have liberal accounting often have weak operating and financial characteristics, some believe this market activity reflects the underlying corporate economics rather than the accounting practices. This belief is consistent with the efficient market theories.

Finally, it should be noted that the research to date has not been able to show satisfactorily the relationship between accounting information, stock prices, and investment returns.

Critical Skill

An understanding of financial analysis is essential to those who use and issue corporate financial statements. Issuers who hope to use financial statements to influence others will be more likely to succeed if they know how statement users analyze financial data. On the other hand, financial statement users will be less likely to respond naively to financial data if they know how to analyze it correctly. Through research we are learning more each day about how around the world statement users analyze financial statements and how the results of these analyses are used. Anyone interested in corporate financial reporting should follow efficient market research efforts of financial economists on a current basis, as today’s knowledge in this area is fast being made obsolete.

⁴ Altman’s Z-score is not a reasonable predictor of financial company bankruptcies. It cannot capture the effect of the off-balance sheet financing employed by many financial firms.